

0.1 Section 1.4 - Propositional Functions and Quantifiers

First, let's start with an example

Example 1

Is $x + 3 = 10$ a proposition?

Solution:

No, since x is not known

However, if x is known, then $x + 3 = 10$ is a proposition

The truth value depends on the value of x

Definition 1 (Propositional Function)

We call $x + 3 = 10$ a **propositional function** since the truth value of the proposition depends on the value of x

Example 2 (Examples of Propositional Functions)

For the given propositional functions, determine the truth value for the given value of variable(s)

1. $P(x) : x + 3 = 10$

(a) $P(6)$

Solution:

FALSE since $6 + 3 \neq 10$

(b) $P(7)$

Solution:

TRUE since $7 + 3 = 10$

2. $P(a, b, c) : a^2 + b^2 = c^2$

(a) $P(11, 60, 61)$

Solution:

TRUE

$$11^2 + 60^2 = 121 + 3600 = 3721$$

$$61^2 = (60 + 1)^2 = 3600 + 120 + 1 = 3721$$

(b) $P(147, 201, 513)$

Solution:

FALSE

$$147^2 + 201^2 = \text{odd} + \text{odd} = \text{even}$$

$$513^2 = \text{odd} \neq \text{even}$$

Example 3 (Examples of Notation) 1. Is "there exists a real number x such that $x + 3 = 10$?" a proposition?

Solution:

Yes, and this proposition is TRUE since $x = 7$ works

We can also rewrite this as:

$$\exists x \in \mathbb{R} \mid x + 3 = 10$$

2. Is "for all real numbers x , $x + 3 = 10$ " a proposition?

Solution:

Yes, but the proposition is FALSE since $x = 6$ doesn't work

We can also rewrite this as

$$\forall x \in \mathbb{R} \mid x + 3 = 10$$

Definition 2 (Quantifiers)

The **existential quantifier** $\exists$ means "there exists"

The **universal quantifier** $\forall$ means "for all" or "for every"

Example 4 (Examples of Quantifiers)

Write each proposition using quantifiers, and determine their truth values

1. There exists an integer x such that $x^2 + 2 > 3x$

Solution:

$$\exists x \in \mathbb{Z} \mid x^2 + 2 > 3x$$

TRUE since $x = 3$ gives $9 + 2 > 9$

2. For all real numbers x , $x^2 + 2 > 3x$

Solution:

$$\forall x \in \mathbb{R} \mid x^2 + 2 > 3x$$

FALSE since $x = 1$ gives $1 + 2 > 3$

Note

We only need one counter example to disprove a proposition that has $\forall$

3. There exists an integer x such that $x^3 = 31687$

Solution:

$$\exists x \in \mathbb{Z} \mid x^3 = 31687$$

FALSE since $\sqrt[3]{31687} \notin \mathbb{Z}$

Note

We needed to verify that every integer does not satisfy the proposition in order to prove it

The negations of each quantifier are as follows:

$$\neg \forall \equiv \exists$$

$$\neg \exists \equiv \forall$$

Example 5 (Negation of Quantifiers)

Give the negation of the following propositions:

1. $\forall x \in \mathbb{R} \mid x^2 + 1 > 0$ **TRUE**

Solution:

$\exists x \in \mathbb{R} \mid x^2 + 1 \leq 0$ **FALSE**

2. $\exists a \in \mathbb{Z} \mid a^3 + 2 = 3a$ **TRUE** ($a = 1$)

Solution:

$\forall a \in \mathbb{Z} \mid a^3 + 2 \neq 3a$ **FALSE**

3. For every real number x , $x^2 + 30 < 11x$ **FALSE** ($x = 1$)

Solution:

$\exists x \in \mathbb{R} \mid x^2 + 30 \geq 11x$ **TRUE**

4. For every prime integer $p > 3$, $p^2 - 1$ is divisible by 24 **TRUE**

Solution:

There exists a prime integer $p > 3$, $p^2 - 1$ is not divisible by 24 **FALSE**

Work

$$p^2 - 1$$

We want $p^2 - 1$ divisible by both 8 and 3

$$(p^2 - 1) = (p - 1)(p + 1)$$

$$(p - 1)(p + 1) = (\text{even})(\text{even})$$

- p is prime
- $p > 3$
- p is odd

Example 6

Statement	When TRUE	When FALSE
$\forall x, P(x)$	$P(x)$ is TRUE for all x	There exists at least one x such that $P(x)$ is FALSE
$\exists x, P(x)$	There exists at least one x such that $P(x)$ is TRUE	For all x such that $P(x)$ is FALSE

Example 7 (Examples with Negation)

Statement	Equivalent	When TRUE	When FALSE
$\neg \forall x, P(x)$	$\exists x, \neg P(x)$	There is at least one x such that $P(x)$ is FALSE	For all x , $P(x)$ is TRUE
$\neg \exists x, P(x)$	$\forall x, \neg P(x)$	$P(x)$ is FALSE for all x	There exists at least one x such that $P(x)$ is TRUE

Example 8 (Examples of Rewriting with Quantifiers)

Rewrite the following with Quantifiers

1. For all positive integers, either $6n + 1$ or $6n - 1$ is prime

Solution:

$\forall n \in \mathbb{Z}^+, 6n - 1$ is prime $\vee 6n + 1$ is prime

FALSE since if $n = 20$ then $6n - 1 = 119$ and $6n + 1 = 121$ are **composite** (not prime)

2. There exists an integer n such that $n^2 + n + 41$ is composite

Solution:

$\exists n \in \mathbb{Z}, n^2 + n + 41$ is composite

TRUE since if $n = 41$ then

$$\begin{aligned} n^2 + n + 41 &= 41^2 + 41 + 41 \\ &= 41(41 + 1 + 1) \end{aligned}$$

which is composite